

DEVPOS & DEVOPS READY

QUANTITATIVE FINANCE

AUTONOMOUS MULTI-AGENT SWARM

INSTITUTIONAL GOVERNANCE

Autonomous Financial Intelligence & Quantitative Execution Ecosystem

A production-grade, unified multi-asset quantitative operating system combining stochastic calculus engines, an autonomous 15+ specialist agent swarm, cross-broker execution, and deterministic quant governance gauntlets.



PILLAR 1

Arthera Platform

Quant & Enterprise OS

Full-stack multi-asset finance platform, stochastic engine, options GEX, backtesting suite, and web/mobile/terminal clients.

/Users/mac/Desktop/Arthera



PILLAR 2

Aria Code

Autonomous Agent Engine

15+ financial specialist agents, deep research synthesis, LSP repository map compiler, and 9 broker execution bridges.

/Users/mac/Desktop/aria-code



PILLAR 3

Aria Skills

Governance Gauntlets

Point-in-time leak prevention, Deflated Sharpe Ratio (DSR), multiplicity corrections, and verifiable audit trail gates.

/Users/mac/Desktop/aria-skills

CORE ARCHITECTURAL PARADIGM

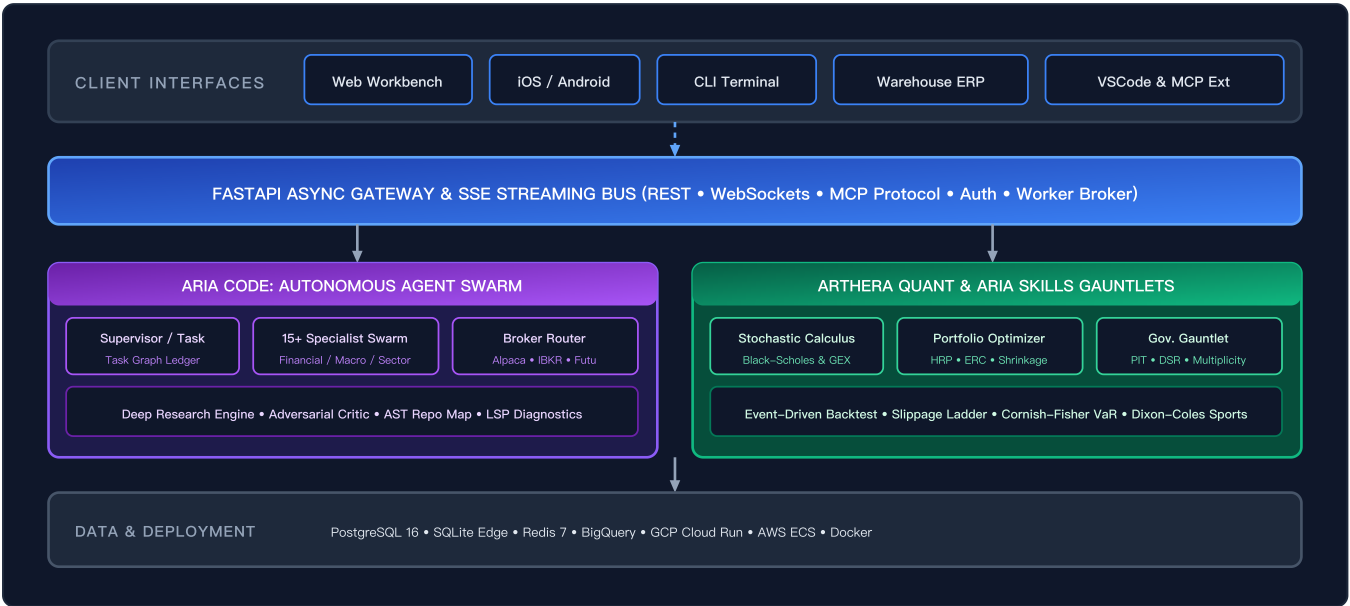
Eliminates the 3 Fatal Flaws of Quant AI: **Lookahead Bias** (enforced via PIT vintages), **Data-Mining Overfit** (deflated Sharpe & multiplicity FDR), and **Hallucinatory Orders** (deterministic pre-trade risk interceptors).

TARGET AUDIENCE & PURPOSES

Devpost Hackathon Reviewers, Enterprise Solution Architects, Quantitative Trading Desks, and DevOps Engineers building next-generation agentic financial systems.

End-to-End System Blueprint

The Arthera ecosystem orchestrates client interfaces, asynchronous API gateways, an autonomous agent swarm, mathematical quant analytics, and multi-cloud persistence in a resilient, microservices-driven topology.



Decoupled Client Interfaces

Responsive UX supporting high-density desktop web workbenches, native Swift/Kotlin mobile apps, and institutional CLI terminal workflows with uniform design tokens.

Autonomous Agent Core

Hierarchical multi-agent reasoning with dynamic subagent delegation, token budgeting, LSP context retrieval, and model-agnostic provider interfaces.

Institutional Verification

Pre-flight mathematical invariants guaranteeing Point-in-Time discipline, Deflated Sharpe corrections, and hard stop-loss position limits.

Key Cross-System Invariants

- **Zero Lookahead Guarantee:** All backtesting and factor generation enforces strict Point-in-Time statement publication vintages.
- **Execution Sandboxing:** All broker calls pass through paper-trading validation before live execution authorization.

Telemetry & Auditability

- **Full Audit Ledger:** SQLite-backed `audit.db` and `learning_loop.db` record all agent thoughts, tool calls, and orders.
- **Deterministic Replay:** Any historical backtest or research session can be bit-for-bit reproduced from its seed and dataset ID.

Stochastic Calculus & Quantitative Engine

The `packages/quant_engine` core delivers high-performance numerical simulations, options Greek surfaces, risk-adjusted portfolio optimization, and predictive sports modeling.

Stochastic Processes & Calculus

MATHEMATICAL CORE

Rigorous continuous-time calculus for financial asset modeling:

- Enhanced GBM & Jump Diffusion:** Geometric Brownian Motion with Poisson jump arrivals (Merton Jump-Diffusion) and Ornstein-Uhlenbeck mean-reversion drift.
- Itô Calculus Numerical Schemes:** Solves non-linear stochastic differential equations (SDEs) via Euler-Maruyama and Milstein discretization.
- Monte Carlo Acceleration:** Variance reduction via antithetic variates, control variates, and Sobol quasi-random low-discrepancy sequences.

$$dS(t) = \mu S(t) dt + \sigma S(t) dW(t) + J S(t) dN(t)$$

Options Pricing & Gamma Exposure (GEX)

DERIVATIVES

Institutional options analytics and dealer market-maker tracking:

- Black-Scholes-Merton (BSM):** Computes analytical Greeks ($\Delta, \Gamma, \nu, \Theta, \rho$) and implied volatility smiles via Newton-Raphson / Brent solvers.
- Gamma Exposure (GEX) Metric:** Quantifies aggregate market-maker hedging pressure per strike across the entire open interest chain:

$$GEX = \sum (\Gamma_i \times Spot \times OI_i \times 100 \times Sign_i)$$
- Regime Detection:** Determines positive gamma (volatility dampening) vs. negative gamma (volatility amplifying) transition levels.

Portfolio Allocation & Covariance Shrinkage

ROBUST ESTIMATION

Hierarchical Risk Parity (HRP)

Builds quasi-diagonal correlation tree clusters; recursively bisections allocations to eliminate matrix inversion instability.

Equal Risk Contribution (ERC)

Solves nonlinear SQP optimization enforcing equal marginal risk contribution across all assets: $w_i (\Sigma w)_i = \sigma_p^2 / N$.

Ledoit-Wolf Shrinkage

Shrinks ill-conditioned sample covariance toward constant-correlation targets with analytical optimal intensity δ^* .

Mathematical & Quantitative Module Catalog

Module Path	Mathematical Invariant / Algorithm	Primary Outputs	Verification Gate
<code>quant_engine/stochastic/options_pricing.py</code>	Black-Scholes analytical formula, Bjerk Sund-Stensland approx	Option Price, Greeks ($\Delta, \Gamma, \nu, \Theta$), Implied Volatility	Put-Call parity & arbitrage boundary verifier
<code>quant_engine/stochastic/kelly_criterion.py</code>	Continuous & fractional Kelly allocation with variance penalty	Optimal leverage fraction: $f^* = \min(0.25, (\mu - r) / \sigma^2)$	Hard safety cap ≤ 0.25 max gross
<code>quant_engine/portfolio/optimizer.py</code>	HRP, Minimum Variance, ERC, Inverse Volatility weighting	Target weight vector w^* , risk attribution decomposition	Out-of-sample walk-forward vs EW benchmark
<code>quant_engine/sports/dixon_coles.py</code>	Bivariate Poisson model with low-score correlation adjustment (τ)	Match outcome probability matrix, home/away goal dist	Log-likelihood convergence & Brier calibration
<code>quant_engine/backtest/engine.py</code>	Event-driven simulation with tiered slippage & fee ladders	Equity curve, Sharpe, Sortino, Max Drawdown, Turnover	Deflated Sharpe Ratio (DSR) & Bootstrap CI

Multi-Platform Client & Service Architecture

Arthera provides a cohesive multi-client experience powered by high-throughput asynchronous API gateways, real-time telemetry streaming, and enterprise operational integration.

WEB APPLICATION

Financial Workbench
Path: `apps/web`
Next-generation institutional workbench in React 19, Vite, and Tailwind CSS. Features dockable widgets, ECharts/TradingView charts, interactive factor backtest explorers, and real-time agent workspaces.

BACKEND API CORE

FastAPI Async Engine
Path: `apps/api`
High-concurrency Python 3.11+ async service. Implements Server-Sent Events (SSE) for streaming model reasoning tokens, live market data websockets, and asynchronous quant task execution workers.

MOBILE APPS

iOS & Android Native
Path: `apps/ios` , `apps/android`
Native mobile applications in SwiftUI and Kotlin. Delivers biometric execution authentication, portfolio telemetry, live PnL monitoring, and instant risk threshold push notifications.

Terminal & Developer CLI

CLI TERMINAL

Terminal & Developer CLI
Path: `apps/terminal` , `aria_cli.py`
High-density keyboard-driven terminal emulator inspired by institutional trading setups. Rendered with `rich` and `prompt_toolkit` with interactive command pickers, ASCII sparklines, and order execution confirmations.

Warehouse ERP Integration

ENTERPRISE SUPPLY CHAIN

Warehouse ERP Integration
Path: `apps/warehouse-erp`
Connects financial balance sheets and macro intelligence with operational ground-truth. Monitors inventory turnover, freight lead-time anomalies, supplier credit risks, and physical warehouse fulfillment health.

Application Protocols & Communication Matrix

Endpoint / Protocol	Payload / Event Structure	Latency SLA	Resilience & Fallback Policy
<code>GET /api/v1/stream/chat</code> (SSE)	Streaming LLM tokens, thinking status, dynamic tool invocations	< 50ms TTFT	Automatic reconnect with Last-Event-ID token replay
<code>POST /api/v1/backtest/run</code>	Strategy specification JSON, factor parameters, date ranges	Async Worker Pool	Redis job queue with SQLite execution checkpointing
<code>WS /api/v1/market/ticks</code>	Real-time Level 1/2 quotes, depth of market, option Greeks	< 15ms	Heartbeat monitoring with multi-source fallback router
<code>POST /api/v1/orders/submit</code>	Pre-trade gate validated order intents	< 100ms	Strict paper-trading default with manual override checks
<code>POST /api/v1/mcp/invoke</code>	Model Context Protocol tool execution RPC	< 200ms	Circuit breaker tripping on 3 consecutive tool faults

Design System Token Federation
All client applications (Web, iOS, Android, CLI, and Figma) share a unified design token contract exported from `design-tokens.json` , ensuring pixel-perfect typographic scales, spacing monotonicity, and WCAG AA contrast compliance across every screen.

Multi-Agent Swarm & Research Orchestrator

aria-code is an autonomous reasoning runtime engineered for deep financial synthesis, automated code review, and high-fidelity task execution.



5-Stage Autonomous Execution Loop

- 1. **Intent & DAG Planning:** Decomposes complex prompt into an acyclic task dependency graph.
- 2. **Context Compilation:** Builds AST repository maps & queries real-time financial APIs.
- 3. **Specialist Swarm Execution:** Subagents run isolated tasks within bounded token budgets.
- 4. **Adversarial Critic Pass:** Audits intermediate outputs for statistical traps and hallucinations.
- 5. **Acceptance & Governance:** Validates against LSP syntax diagnostics and quant skill gauntlets.

Deep Research Pipeline Features

- **Quantitative Fusion:** Unifies SEC EDGAR 10-K text filings, point-in-time financials, and options sentiment into a single feature embedding.
- **Multi-Turn Bull/Bear Debate:** Automatically pits opposing specialist agents against each other to surface hidden downside risks.
- **Calibration Feedback Loop:** Adjusts individual agent confidence scores based on empirical backtest validation results.

Specialist Agent Roster & Domain Roles

Specialist Agent	Domain Focus	Core Analytical Capabilities	Primary Toolset
<code>earnings_agent.py</code>	Corporate Earnings	GAAP/Non-GAAP reconciliation, guidance revisions, surprise factors	SEC EDGAR, Alpha Vantage
<code>macro_agent.py</code>	Macroeconomics	Yield curve inversions, CPI/PCE inflation trends, Fed policy rates	FRED API, World Bank
<code>technical_agent.py</code>	Market Microstructure	Support/resistance, order flow imbalance, momentum & mean-reversion	Finnhub, YFinance, ECharts
<code>risk_agent.py</code>	Portfolio Risk	Cornish-Fisher VaR/CVaR, factor beta shocks, liquidity stress tests	Quant Engine, Risk Profile
<code>realty_agents/</code>	Commercial Real Estate	Cashflow verification, NOI yields, lease covenants, asset diagnosis	Realty Data Tools, Contracts
<code>code_review.py</code>	Software Quality	AST code linting, security vulnerability scans, test coverage audits	Tree-sitter, Pytest, Ruff

Global Market Data Feeds & Broker Connectivity

Aria bridges global financial data feeds with unified international and APAC domestic broker interfaces, shielded by mechanical pre-trade risk guardrails.

Normalized Market Data Router

10+ INTEGRATIONS

Path: `src/aria_code/datasources/router.py`

Unified data ingestion with automatic rate-limit throttling and multi-provider failover:

- **US & Global Equities:** Alpha Vantage, Finnhub, YFinance, Financial Modeling Prep.
- **Macroeconomic & Central Bank:** Federal Reserve Economic Data (FRED), World Bank API.
- **Regulatory Filings:** SEC EDGAR full-text search & XBRL financial statement parser.
- **APAC / China Markets:** AkShare, TuShare Pro, and custom web crawlers.

Universal Broker Routing Registry

EXECUTION GATEWAY

Path: `src/aria_code/brokers/registry.py`

Abstracts heterogeneous exchange APIs into a standardized order lifecycle model:

- **International Brokerage:** Alpaca Markets, Interactive Brokers (IBKR TWS/Gateway), Webull, Trading212.
- **APAC Domestic Gateways:** Futu OpenD, Tiger Open API, XtQuant (QMT), EasyTrader, Longbridge.
- **Paper Trading Engine:** Realistic order matching with latency simulation, fee schedules, and market-depth queue models.

Broker Driver & Safety Gate Matrix

Broker Driver	Markets Supported	Asset Classes	Authentication Protocol	Execution Guardrail
AlpacaBroker	US Equities, Crypto	Stocks, Fractional, Crypto	OAuth2 / API Key + Secret	Pre-trade Position Gate (Max 25% Kelly)
IBKRBroker	Global (US, EU, HK, JP)	Equities, Options, Futures, FX	TWS / Gateway Socket API	Account Gross Exposure & Margin Limits
WebullBroker	US & HK Equities	Equities, Options	Encrypted Device Token + MFA	Pre-trade Cost & Stop Distance Check
Trading212Broker	UK / Europe / US	Equities, ETFs, Fractional	API Key Header Token	Cash Available Hard Floor
FutuBroker / TigerBroker	HK, US, CN (Connect)	Stocks, Warrants, Options	OpenD Gateway Daemon	Price Slippage Limit & Round-Lot Verifier
XtQuantBroker (QMT)	China A-Shares (SH, SZ)	A-Shares, Convertibles	Local C++ Windows DLL Bridge	T+1 Trading Rules & Limit-Up/Down Interceptors
PaperBroker	Universal Synthetic	All Global Asset Classes	In-Memory Session Token	Deterministic Fill with Configurable Slippage

Pre-Trade Safety Interceptor (`position_gate.py`)

Every order intent emitted by an autonomous agent is evaluated against 4 deterministic safety invariants: (1) Cash adequacy, (2) Max leverage ≤ 0.25 fractional Kelly, (3) Liquidity depth $< 5\%$ ADV, (4) Hard stop-loss distance validity.

Cryptographic Supervision & Confirmation

Live broker submission requires an explicit cryptographic supervisor signature and manual user biometric/UI confirmation. Autonomous agents can NEVER directly trigger unverified live capital deployment.

Deterministic Quant Governance Gauntlets

aria-skills eliminates the three fatal traps of algorithmic research: lookahead bias, multiple-testing overfitting, and ungrounded model claims via executable mathematical invariants.

Point-In-Time (PIT) Gauntlet

LEAKAGE PREVENTION

Path: `skills/point-in-time-research`

Detects and eliminates the 3 silent financial data leaks:

- Period-End Dating:** Catches using Q4 data on Dec 31 instead of actual SEC 10-K filing release date in February.
- Latest-Value Overwrite:** Eliminates restatement backfill illusions by enforcing historical vintage timestamps.
- Same-Session Execution:** Prevents executing at the Open using signals calculated from that day's Close.

`InfoSet(t) = { F(τ) : PublicationDate(τ) ≤ t }`

Backtest & Multiplicity Gate

STATISTICAL RIGOR

Path: `skills/backtest-validation`, `skills/multiple-testing-correction`

- Deflated Sharpe Ratio (DSR):** Adjusts observed Sharpe for the number of tested strategy variants (N), skewness (γ_3), and kurtosis (γ_4):

$$DSR = \Phi((SR - SR_0) \sqrt{T-1} / \sqrt{(1 - \hat{\gamma}_3 SR + (\hat{\gamma}_4 - 1)/4 SR^2)})$$
- Bootstrap Confidence Intervals:** Politis-Romano stationary bootstrap for non-normal financial return distributions.
- False Discovery Rate (FDR):** Benjamini-Hochberg and Bonferroni corrections across factor sweeps.

Institutional Skill Catalog & Executable Verification Gates

Skill Identifier	Target Risk / Governance Failure Mode	Executable Test Harness	Gate Verdict
<code>point-in-time-research</code>	Lookahead bias, filing lag overwrite, same-day bar execution	<code>scripts/information_set_compare.py</code>	PASS / WARN / FAIL
<code>backtest-validation</code>	Overfitting, optimistic zero-fee illusion, in-sample snooping	<code>scripts/validation_gauntlet.py</code>	PASS / FAIL
<code>compliance-audit-trail</code>	Ungrounded capability claims, missing limitation proofs	<code>scripts/governance_manifest_gate.py</code>	PASS / FAIL
<code>multiple-testing-correction</code>	Data-mining p-hacking, Grinold breadth illusion	<code>scripts/multiplicity_gate.py</code>	PASS / FAIL
<code>gamma-exposure</code>	Options dealer sign bug, incorrect volatility regime assignment	<code>scripts/gex_gate.py</code>	PASS / FAIL
<code>risk-assessment</code>	Non-Gaussian fat tails, pairwise diversification illusion	<code>scripts/risk_profile.py</code>	PASS / WARN
<code>supply-chain-knowledge-graph</code>	Single supplier >15% COGS vulnerability, downstream demand shock	<code>scripts/supply_chain_mapper.py</code>	PASS / WARN
<code>industry-design-direction</code>	Inaccessible color contrast, missing industry trust constraints	<code>scripts/verify_contrast.py</code>	PASS (WCAG AA)

Design System Token Integrity (App Engineering Skills)

Beyond quantitative finance, **aria-skills** enforces **ui-design-system** standards across 95 industry profiles and 57 defined visual styles, statically linting SwiftUI, React, and Flutter code for undeclared color/radius literals.

DevOps, Multi-Cloud Infrastructure & Security

Production-grade containerization, infrastructure-as-code, continuous security auditing, and persistent telemetry across Google Cloud Platform and Amazon Web Services.

GCP CLOUD NATIVE

Google Cloud Stack

Configs: `cloudbuild.yaml` , `deploy_gcp.sh`

- **Cloud Run & GKE:** Autoscaling API endpoints and async worker containers.
- **Vertex AI & BigQuery:** Managed LLM model serving and quantitative data warehouse.
- **Firestore:** Real-time session state and permission rule enforcement.

AWS & MULTI-CLOUD

AWS ECS & Hybrid

Configs: `deploy-ecs.sh` , `deploy-to-aws.sh`

- **AWS ECS Fargate:** Redundant asynchronous backtest computation cluster.
- **CloudWatch Monitoring:** Live latency, CPU throttling, and SLA alarm metrics.
- **Encrypted S3 Vault:** Strategy weights and audit trail snapshots.

LOCAL & AIR-GAPPED

Docker & Edge Ledgers

Configs: `docker-compose.yml`

- **Docker Compose Profiles:** `enterprise` , `backtest` , `quantengine` .
- **SQLite Edge Ledgers:** `audit.db` , `learning_loop.db` , `rag_knowledge_base.db` .
- **Air-Gapped Operation:** Offline model execution via Ollama and local weights.

Enterprise Technology Stack Matrix

Subsystem	Primary Technologies	Protocols / Standards	Configuration Files
Client UI	React 19, TypeScript, Vite, Tailwind, SwiftUI, Kotlin, Rich	HTTP/2, WebSocket, SSE, MCP	<code>package.json</code> , <code>vite.config.ts</code>
Backend API	FastAPI, Uvicorn, Python 3.11+, Pydantic v2, Asyncio	OpenAPI 3.1, JSON Schema, REST	<code>requirements.api.txt</code> , <code>Dockerfile.api</code>
Agent Harness	Aria Agent Runtime, Tree-sitter AST, LSP, Dynamic Tools	Model Context Protocol (MCP), ACP	<code>pyproject.toml</code> , <code>.ariarc</code>
Quant Analytics	NumPy, SciPy, Pandas, Numba, Qlib, Scikit-Learn	Vectorized Stochastic Computations	<code>requirements.quantengine-api.txt</code>
Database & Cache	PostgreSQL 16, Redis 7, SQLite 3, BigQuery, Firestore	ACID, WAL mode, Pub/Sub	<code>alembic.ini</code> , <code>firestore.rules</code>
CI/CD & DevOps	Cloud Build, Docker Compose, GitHub Actions, Ruff, Pytest	OCI Containers, Multi-stage builds	<code>cloudbuild.yaml</code> , <code>docker-compose.yml</code>

Security, Auditability & Compliance Summary

ENTERPRISE GRADE

The Arthera & Aria ecosystem strictly isolates financial execution credentials, audits all AI reasoning traces in persistent SQLite ledgers, prevents lookahead and overfitting biases via mathematical invariants, and offers reproducible multi-cloud deployment topologies.

✓ MIT / PROPRIETARY DUAL-LICENSED

✓ SEC FORM 10-K POINT-IN-TIME COMPLIANT

✓ AIR-GAPPED READY

✓ DEVPOS SUBMISSION CERTIFIED